

AirFly Insights: Operational Analysis of Airline Performance.

Data-Driven Visualizations to Enhance Reliability and
Efficiency.

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Executive Summary

Key Finding 1: Market Concentration

Flight activity is highly concentrated; Southwest, Delta, and American lead in volume.

Key Finding 2: The "Morning Advantage"

Operational efficiency is highest in early morning windows (0-6 AM) before cumulative delays cascade.

Key Finding 3: Primary Disruption Drivers

Weather is the primary driver of cancellations (winter months), while "Airline/Carrier" issues dominate internal operational disruptions.

Market Overview:

Volume Analysis

01

Leaders

Southwest, Delta, and American Airlines lead in flight volume.

02

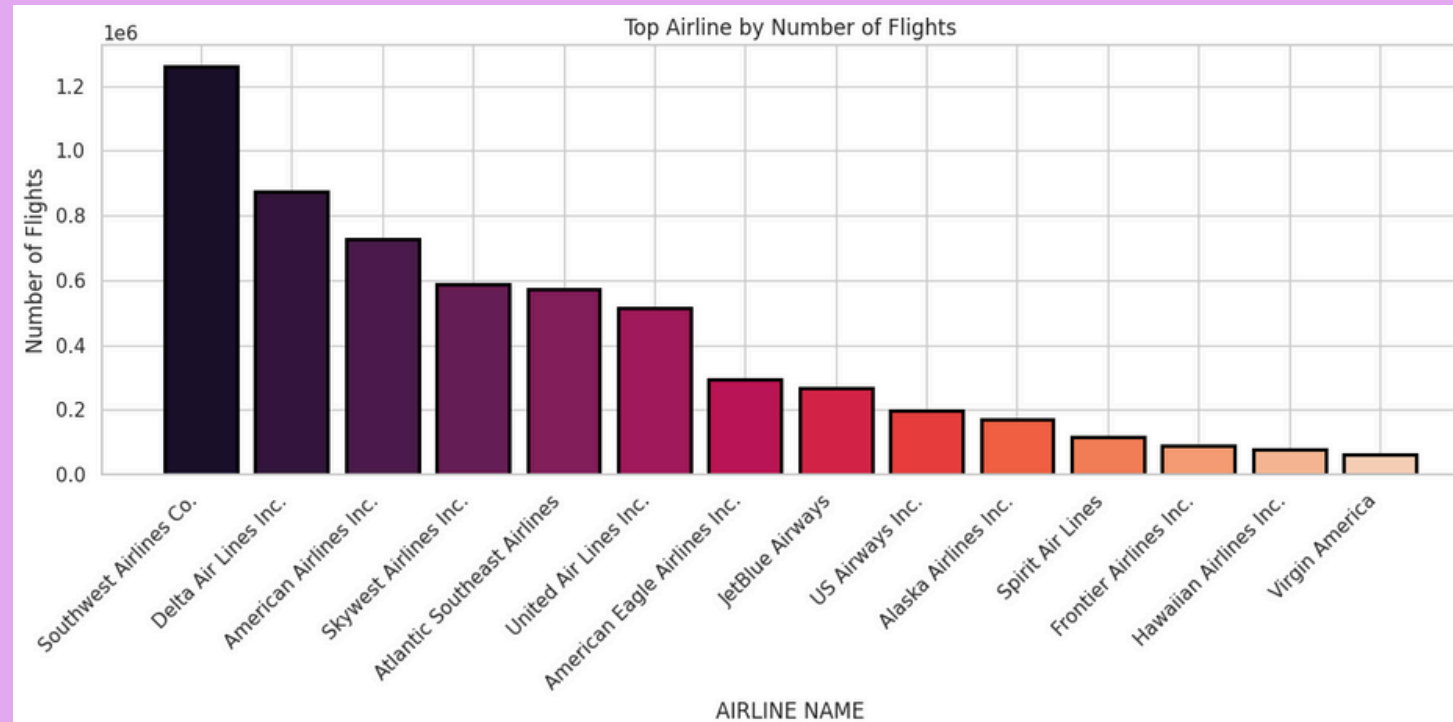
Mid-Tier

Skywest, Atlantic Southeast, and United also have significant operations.

03

Smaller

Frontier and Allegiant operate fewer flights.



Route Density Analysis

01

High Traffic

SFO-LAX, LAX-SFO, JFK-LAX, and LAX-JFK are the most frequent.

02

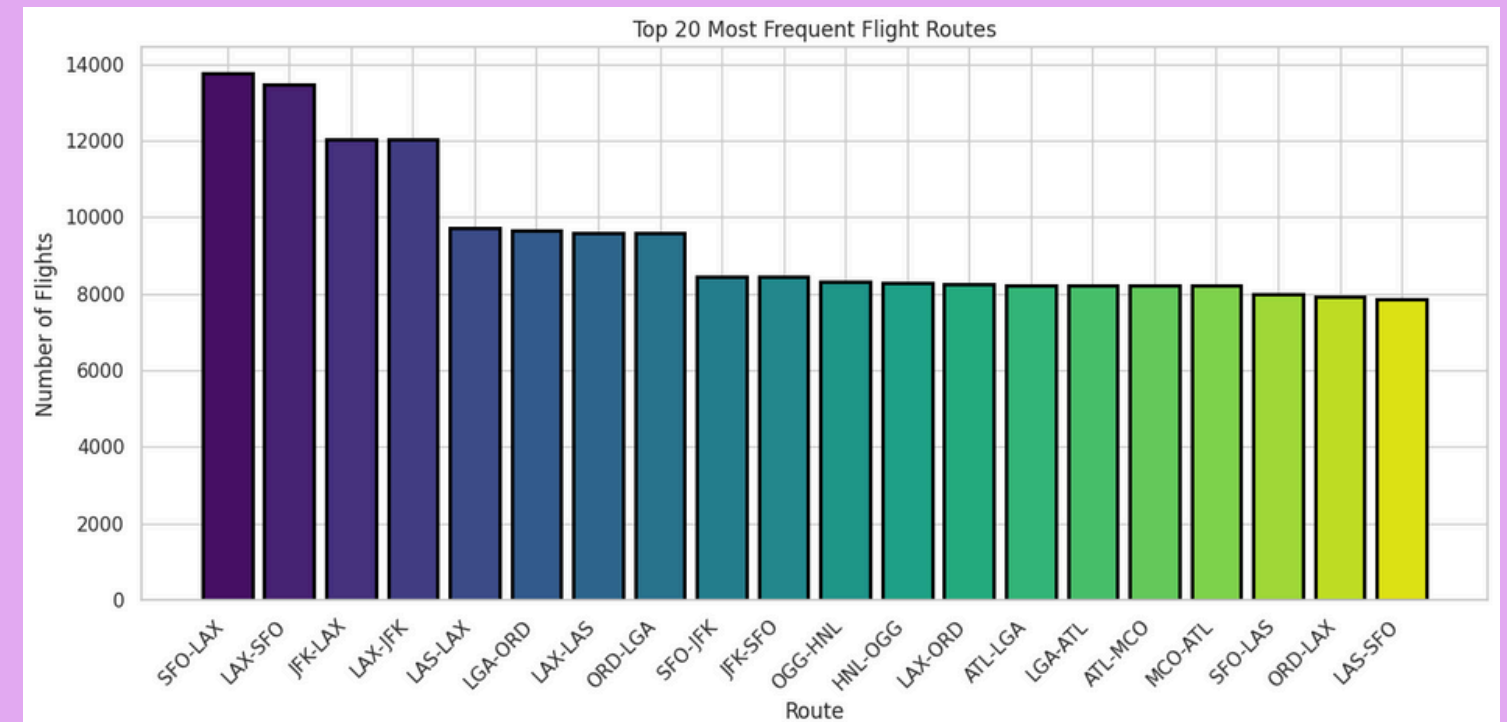
Hub Connections

Many routes connect major hubs (e.g., ORD, ATL, LGA).

03

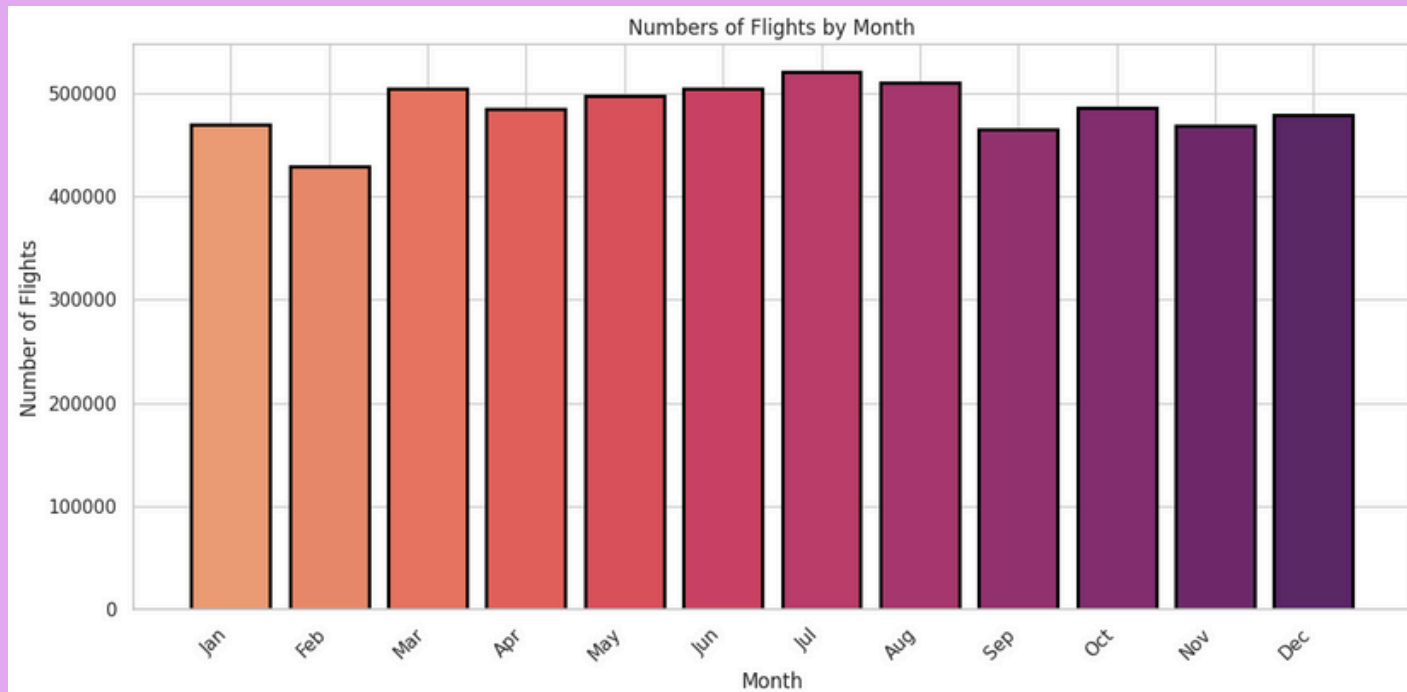
Balanced Demand

Top routes often appear in both directions.



Seasonal & Weekly Demand Patterns

Monthly Trends



01

Peak Months

July and August show the highest flight volumes (summer travel).

02

Shoulder Months

March, May, and June also have high activity.

03

Low Month

February has the fewest flights.

Weekly Demand Analysis

01

Busiest Days

Thursday, Monday, and Friday show the highest flight volumes, indicating peak business and start-of-weekend travel.

02

Mid-Week Activity

Tuesday and Wednesday maintain strong, consistent flight activity.

03

Weekend Trends

Sunday has significant return travel, while Saturday is typically the quietest day for departures.

04

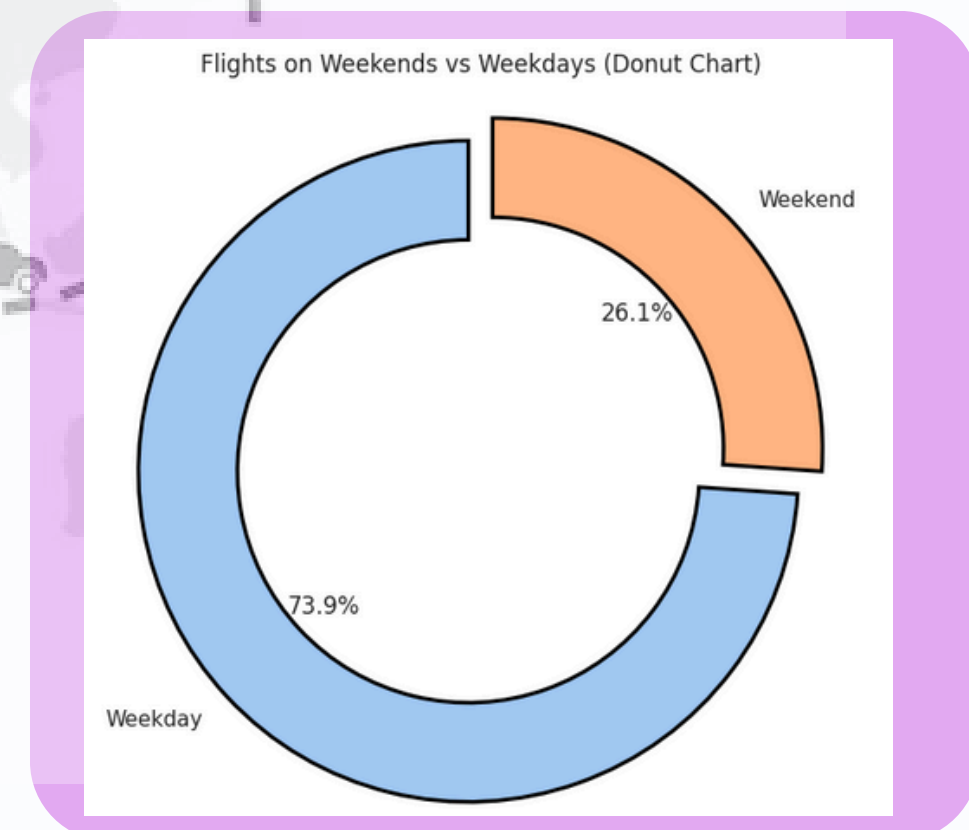
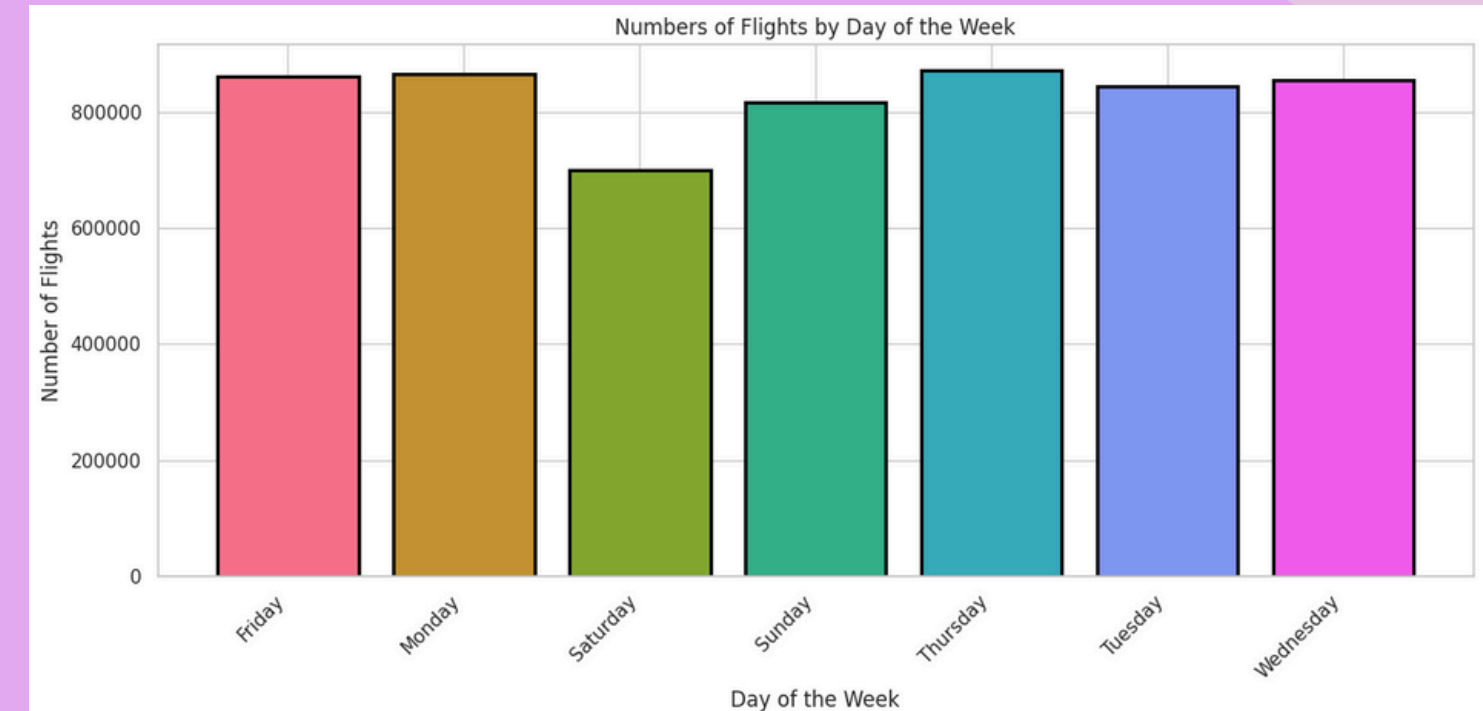
Weekdays Dominant

A large majority of flights (approximately 74%) occur on weekdays.

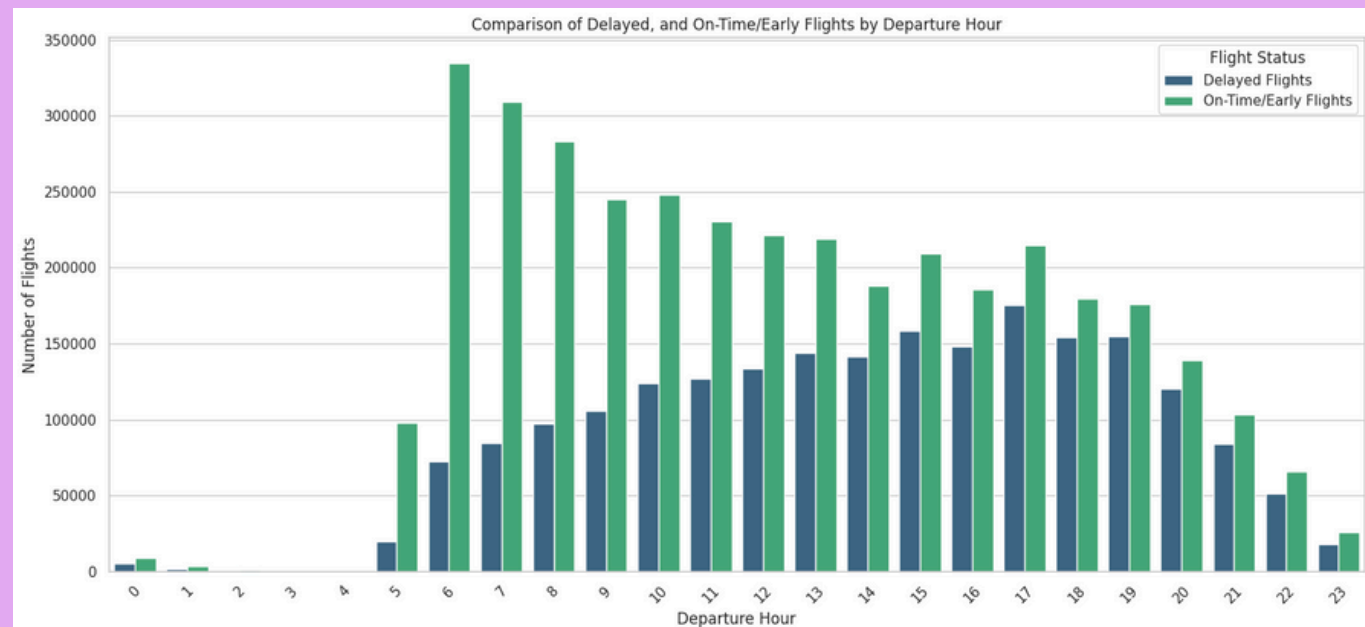
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Weekend Travel

Around 26% of flights happen on weekends.



The "Cumulative Delay" Cascade



01

Peak Delays

Delays (blue bars) are more prominent during the afternoon and evening hours.

02

On-Time Performance

Morning hours (e.g., 6-13) show a higher proportion of on-time/early flights relative to delayed flights.

03

Evening Congestion

As the day progresses, especially in the evening, while total flights remain high, the number and proportion of delayed flights also increase.

Root Cause Analysis – Why Are We Delayed?

Analysis Delay Percentage

Leading Delays

01

Departure and Arrival delays are the most frequent.

Other Significant Factors

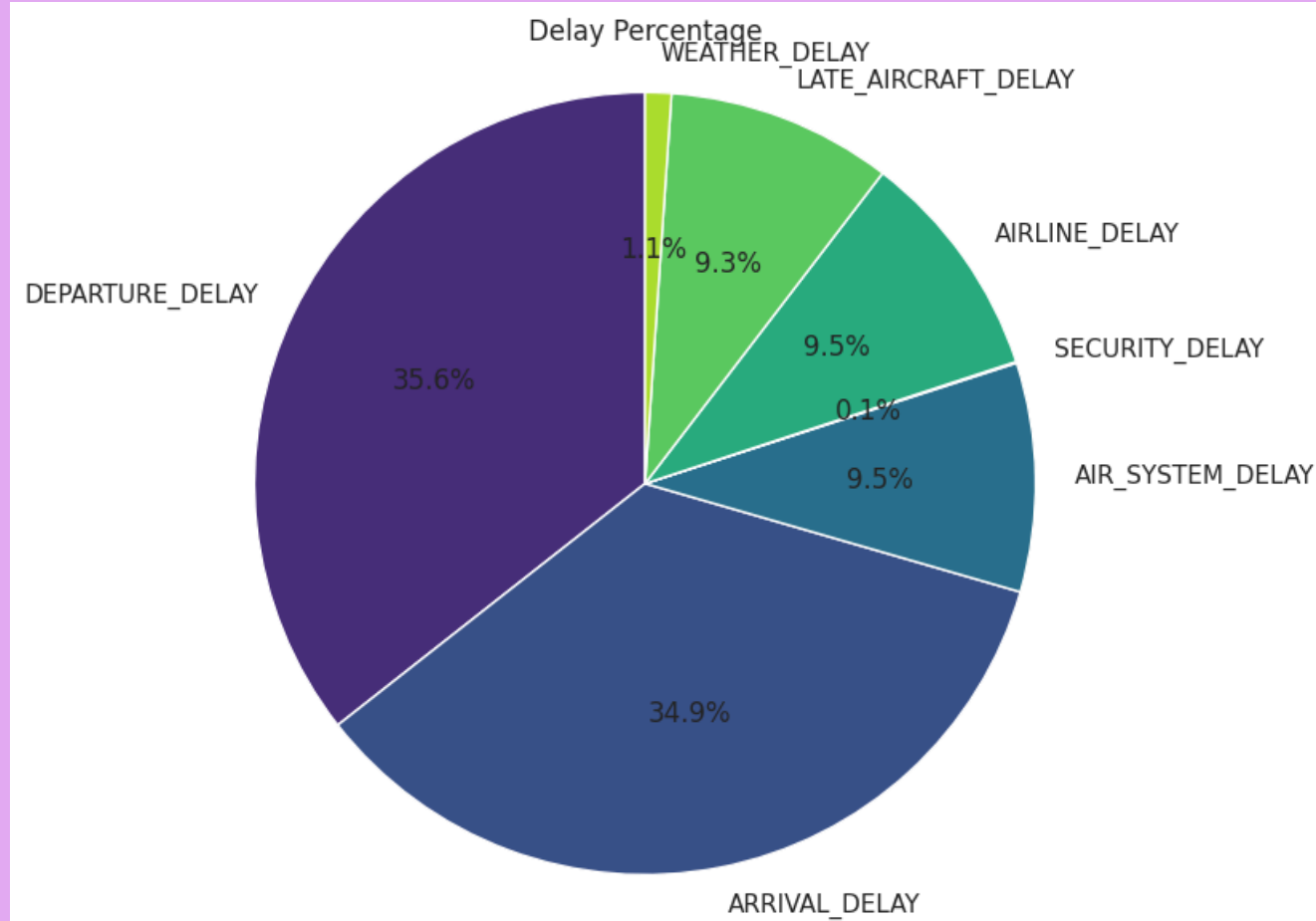
02

Airline, Air System, and Late Aircraft delays also contribute considerably.

Minor Causes

03

Security and Weather delays account for a much smaller percentage.



Analysis of Delay Causes by Airline

01

Airline and Late Aircraft Delays

These are often the largest contributors, indicating internal operational issues or cascading delays.

02

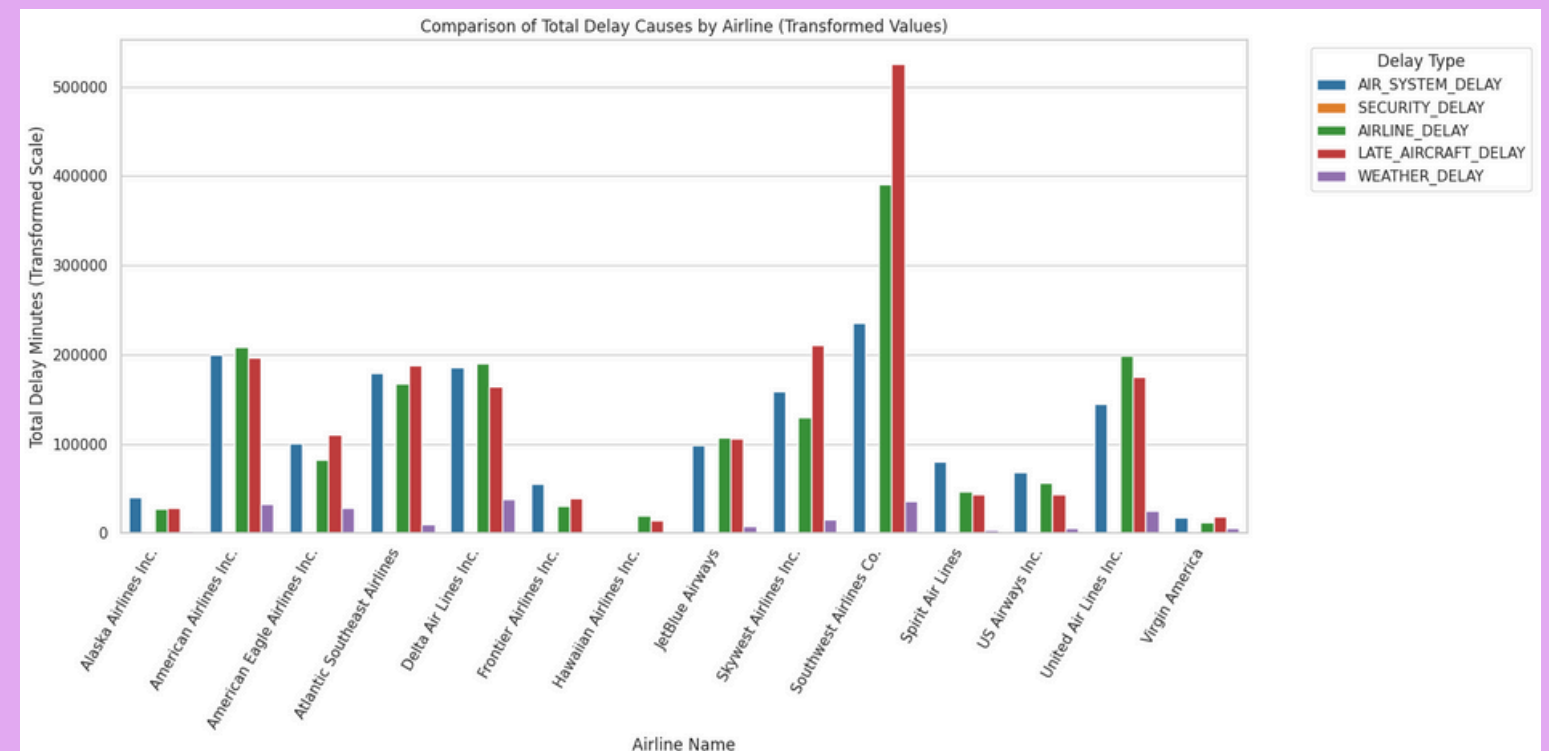
Air System Delays

Also significant, these points point to air traffic control, congestion, or other system-wide issues.

03

Weather and Security Delays

Generally have a smaller impact but can be severe when they occur.



Cancellations Analysis

Monthly Flight Cancellation Trends

Peak Cancellation Periods

01

Cancellations are highest in February, followed by Jan and March, likely due to winter weather.

Low Cancellation Periods

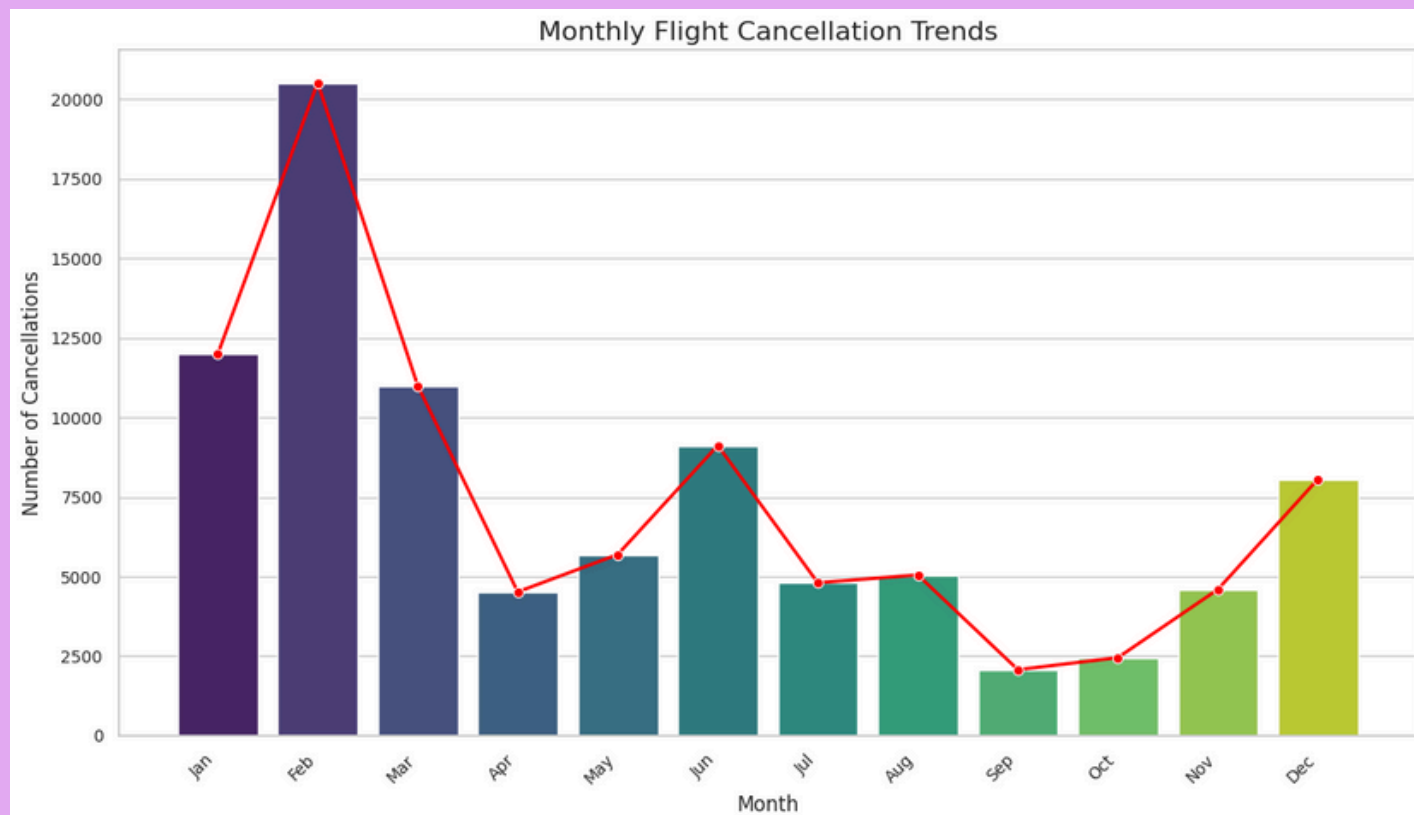
02

The lowest cancellation rates occur from Sep to Dec.

Seasonal Impact

03

Winter/early spring sees the highest flight cancellations; late summer/autumn, the lowest.



Cancellation Reasons

01

Primary Causes

Weather and Airline/Carrier issues are the leading causes of cancellations.

02

Weather Impact

Significant disruptions from storms, fog, etc.

03

Airline Operations

Includes mechanical problems, crew shortages.

04

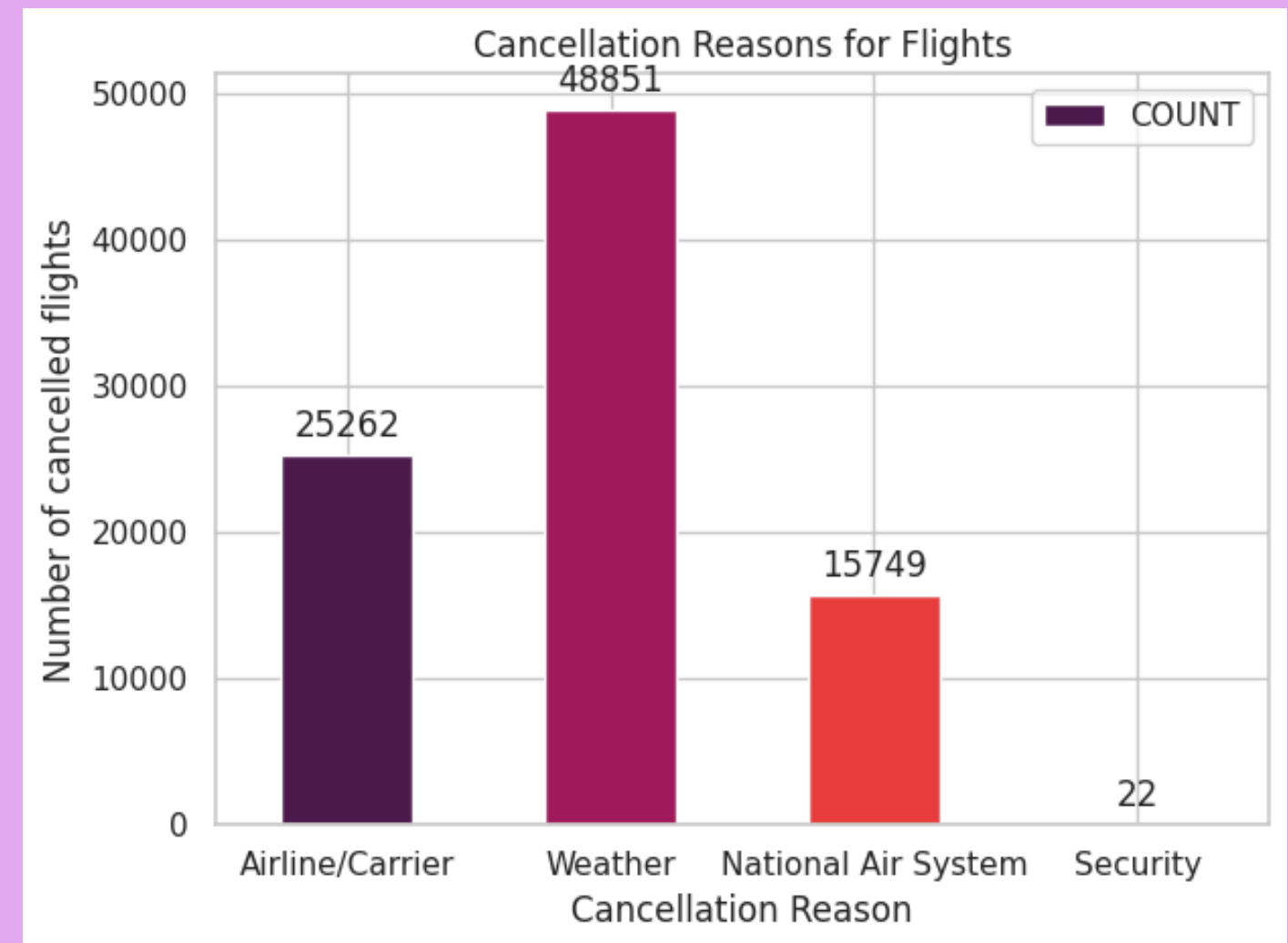
NAS

Relates to air traffic control and airport operations.

05

Security

Accounts for very few cancellations.



Hub Performance & "Hotspot" Map

Size of the Circle

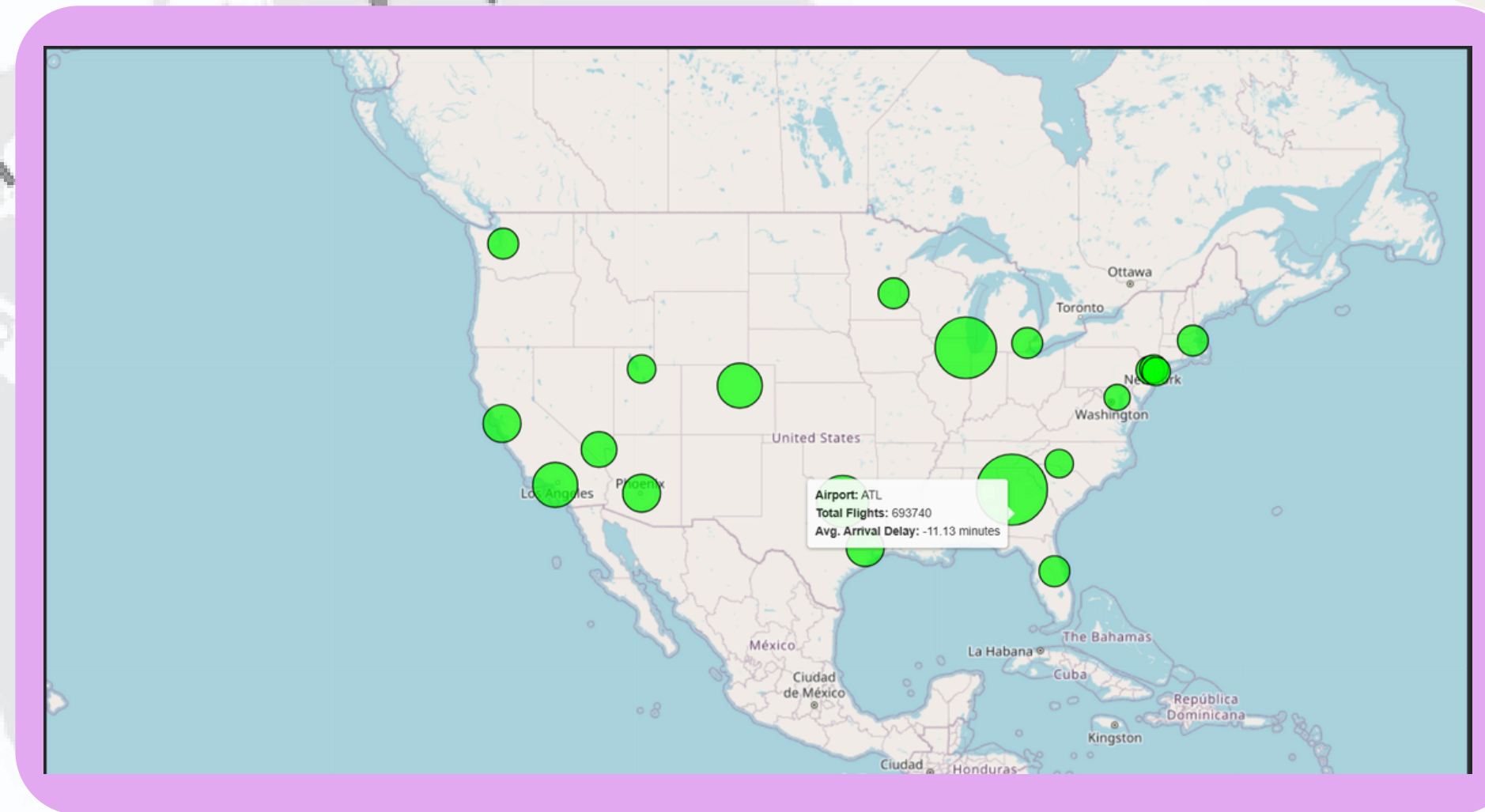
01

Corresponds to the total number of flights handled by the airport (both origin and destination). Larger circles indicate busier airports, highlighting major air traffic hubs.

Color of the Circle

02

Represents the average arrival delay at that airport. The color scale ranges from bright green (very early arrivals) to light green (early), yellow (minor delay/on-time), dark orange (moderate delay), and red (significant delay).



Strategic Recommendations

Maximize the "Morning Efficiency" Window

Prioritize high-value or time-sensitive flights for the 0-6 AM departure block, as this period consistently experiences the lowest average delays across all categories.

Implement Peak-Hour Buffers

Increase scheduling buffers and ground staff during late afternoon and evening blocks to mitigate the cumulative "cascade" effect, where initial disruptions multiply as the day progresses.

Proactive Winter Risk Management

Deploy additional de-icing resources and staff specifically for January and February, as cancellation rates nearly triple during these winter months (2.94%) compared to non-winter periods (1.11%)



Strategic Recommendations

Targeted Hub Interventions

Focus operational review teams on "Hotspot" hubs like Chicago O'Hare (ORD) and Dallas/Fort Worth (DFW), which exhibit higher average delays across airline and air system categories.

Carrier-Specific Operational Audits

Since "Airline/Carrier" issues are a top reason for both delays and cancellations, major carriers should perform internal audits on crew scheduling and mechanical readiness to reduce controllable disruptions.

Seasonal Efficiency Opportunity

Leverage the September to December window for maximizing flight frequency and reliability, as these months show the most stable flying conditions and lowest cancellation rates.





*Any
Questions*

Thank You

